

Hanako Keney

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EDUCATION

Purdue University

West Lafayette, IN

Bachelor of Science in Computer Science, Minor in Mathematics

Expected May 2028

- GPA: 3.6 / 4.0
- Relevant Coursework: Systems Programming, Operating Systems, Computer Architecture, Data Structures & Algorithms, Web App Development, Software Engineering, Linear Algebra, Statistics

EXPERIENCE

CS 193 Head Instructor & Content Developer

May 2025 – Present

Purdue University

West Lafayette, IN

- Designed and delivered curriculum for 500+ students across 3 semesters, covering Git, Linux, Vim, Bash, and collaborative software engineering workflows
- Created 4 lecture slide decks and 2 full homework assignments; delivered 50+ lectures in coordination with the department professor
- Managed and mentored a team of 20+ teaching assistants, coordinating grading, office hours, and course logistics across both academic semesters
- Developed open-source instructional projects on GitHub used as hands-on labs for real-world tooling practice

PROJECTS

Unix Shell Interpreter | *C/C++, Flex, Bison, Unix Systems Programming*

Feb 2026 – May 2026

- Built a fully functional Unix shell from scratch supporting process creation via `fork/exec`, bidirectional pipelines, I/O redirection (`stdout`, `stderr`, `append`), and background execution
- Designed a command parser with Flex and Bison handling complex grammar: quoting, escaping, subshell expansion `$(...)`, and process substitution via named pipes (`mkfifo`)
- Implemented signal handling for `SIGINT` and `SIGCHLD`, zombie process cleanup via `waitpid`, and environment variable management (`setenv/unsetenv/printenv`)
- Developed a raw-mode line editor with left/right cursor movement, history traversal, tab completion, and tilde/wildcard expansion using `regcomp/regexec`

Terminal Portfolio Website | *Vanilla JavaScript, HTML, CSS* | ev3rshade.com

Mar 2026 – Present

- Engineered a browser-based terminal emulator with a virtual filesystem supporting shell commands: `ls`, `cd`, `grep`, `find`, and a read-only Vim-style file viewer
- Implemented command parsing, input history, and tab completion client-side with no backend, mirroring real shell UX in pure JavaScript
- Designed a modular frontend architecture with dynamic content loading, CSS variable theming, and persistent user configuration via `localStorage`

PGSG Grant Application Platform | *React, FastAPI, PostgreSQL, Docker* | GitHub

Nov 2025 – Present

- Containerized the full application stack using Docker, ensuring reproducible environments across development and production deployments
- Designed the PostgreSQL database schema and built a FastAPI backend with an automated initialization pipeline for ingesting grant submission data
- Integrated with Purdue ACM SIGAPP's existing workflows to support live grant application processing

Currents of Culture — Author eBook Platform | *React, Markdown, Stripe*

2025

- Built a full-stack eBook reader and marketing website for a local author, integrating Stripe for payment processing and Markdown rendering for chapter display

TECHNICAL SKILLS

Systems & DevOps: Linux (Bash, shell scripting), Docker, Git, process management, file descriptors, `cron`, `ssh`

Languages: C/C++, Python, JavaScript, SQL, Bash

Frameworks & Tools: React, FastAPI, Flask, Node.js, PostgreSQL, SQLite, pandas, NumPy

Developer Tools: VS Code, GDB, Docker, Neovim

Platforms: GitHub, Brightspace, EdStem